

TEAM PROPOSAL

TEAM NAME	MBK
UNIVERSITY	University of Moratuwa
PRODUCT NAME	BusMate (App)
CATEGORY	Transport

Problem Definition, Analysis, and Solution

Problem Definition

Introduction to Public Transportation in Sri Lanka

With buses being the most popular form of transportation for millions of people, public transportation is essential for daily life in Sri Lanka. The bus system is still the most economical choice for workers, students, and low- to middle-class people. It connects urban centers, suburban communities, and rural locations. Despite this dependence, the system is mainly unofficial: information on bus availability is mostly communicated orally, and schedules are not regularly posted or followed. Buses usually run at the busiest times of the day, but outside of those times, their dependability drastically declines.

Problem Context

A major obstacle that travelers encounter is the uncertainty around bus availability during the night and early morning hours. After a specific hour in the evening, buses stop running on most routes; although passengers are not informed of this clearly. People who are waiting at bus stops are therefore frequently unsure if a bus will ever arrive, which wastes time, leads to frustration, and occasionally raises safety concerns, particularly for those who rely on public transportation for early or late work shifts.

Significance of the Problem

This issue affects society and the economy. It adds stress and uncertainty to commuters' everyday lives. Missed or unclear transportation might result in lower productivity, missed courses, or increased costs for employees and students who depend on ride-hailing services or taxis. On a larger scale, the lack of trustworthy information diminishes confidence in the public transit system and **restricts its potential as a sustainable substitute for private automobiles**. Improving urban mobility, boosting passenger confidence, and aiding Sri Lanka's larger initiatives to update its transportation infrastructure all depend on closing this gap.

Analysis

In 2017, approximately 41% of the transportation in Sri Lanka was based on public transportation, mainly buses and trains, where in urban areas like Colombo, this percentage is observed to be somewhat less (around 30%) compared to rural areas [1]. However, “Statistica”

predicts that the transportation user penetration rate will increase in the next 5 years [2], so we believe that it is high time to move towards sustainable public transportation as soon as possible.

The majority of the population in Sri Lanka is in the low-middle class (based on their income), and a lot of people who are willing to earn an extra income to support the family do night shifts, which require them to transit between their hometown and workplace in early mornings or nights. Since most of these people are from low-income levels, public transportation is the only option for them.

This affects not only those people who work on night shifts but also students and people who have 9-to-5 jobs. The main medium of transportation that these people use is the bus service, which can be either in the private sector or the government sector. However, the bus service in Sri Lanka is identified as quite unpredictable, especially during off-hours.

This inconvenience leads to frustration and affects the personal lives of people as individuals, because this frustration can eventually lead to stress and anxiety. The unpredictability of the arrival of a bus can collapse all the plans that an individual has for the rest of their day. If an individual could know for sure the time of the arrival of the last bus or the location of the next bus, it would save a lot of time and would be very much more convenient.

This also affects the productivity and efficiency of our country, and it is crucial that we act now to address this problem. The need for sustainable cities and communities is also identified by the United Nations under the 17 Sustainable Development Goals released in 2015 [3], and currently, the world is acting towards achieving these goals to build a world where all people can live in peace and prosperity by the year 2030.

Numerous governments worldwide have taken action to increase public transportation's dependability and accessibility. But in Sri Lanka, the bus service hasn't gotten enough attention in this area yet. The web-based train tracking tool [4], which has gained popularity among Sri Lankans, is a noteworthy illustration of progress. With the help of data from the Department of Railways, this system gives customers up-to-date information on train arrivals, departures, and delays, and even sends out notifications for misplaced objects.

In contrast, existing efforts targeting the bus service are limited—most rely on data manually provided by drivers, offering only basic information such as the current location of buses [5].

Another related initiative is Suba Gaman, an app developed by students using GPS. However, it has only been implemented in Kandy, lacks a user-friendly interface, and was not marketed effectively. Our app differs significantly as it not only provides real-time bus location but also incorporates bus speed, seat availability, making it more practical and scalable.

Given that **around 80% of the Sri Lankan population owns a smartphone (ANFREL, 2024)** [6], a well-designed mobile application presents the most effective and scalable solution to address the challenges faced by bus passengers.

But these do not directly approach the problem discussed here. It will be more convenient for the user if they can know the arrival time of the next bus or if any bus will arrive at all on that day and how crowded the bus would be.

Solution

We propose a **smart public transport application- 'BusMate'** that integrates real-time data from bus drivers and AI-driven predictions to provide passengers with accurate, convenient, and trustworthy travel information.

Key information from the driver, such as driver ID, route number, bus number plate, and real-time GPS location, will be gathered by the system. To improve comfort and transparency, cameras will be placed at each bus's entry and departure to monitor passenger counts and show whether there are seats available or whether the vehicle is full.

Our method is unique in that it uses artificial intelligence to determine bus arrival timings based on current route conditions and speed, and whether it is the last bus on the route for the day. Unlike traditional static itineraries, this predictive capability gives travelers more accurate information.

We intend to add an online reservation system to enhance the trip even further. In addition to providing convenience to those who would rather make reservations in advance, this will lower the possibility of bus fare-related fraud by automating the ticketing process.

About a decade ago a card payment system for buses was implemented in Sri Lanka. It failed because card payments were not popular at the time and were poorly marketed. Our solution overcomes these pitfalls by offering a hybrid model passengers can maintain an initial balance (like a season ticket) and fares are deducted seamlessly per trip. This approach ensures secure transactions without exposing bank account details and minimizes the risks.

Passengers will benefit from a more dependable, open, and sustainable public transportation system thanks to this integrated solution. The elimination of uncertainty regarding the arrival time and status of buses would boost system confidence and promote increased use of public transportation.

Product Overview and Uniqueness of the Product

Product Overview

Our app is designed to completely change how people experience bus travel in Sri Lanka. Right now, passengers waste hours at bus stands not knowing when the next bus will arrive, whether it is already full, or even if the last bus has already left. Safety and accountability are also big concerns if you lose something inside a bus, there's no reliable way to trace it back.

With our app, every bus has a simple smartphone inside. That phone powers:

- Live location tracking: passengers see exactly where buses are.
- Seat availability status: full / almost full / seats available.
- Payment system: cashless payments or QR scan, no more savings lost with conductors.
- Bus booking option: reserve a seat in advance, like booking a movie ticket.
- Notifications: "Bus arriving soon", "This is the last bus today."

For passengers, this means saved time, better safety, and reliability. For bus owners, it means no more lost revenue, full visibility of each trip, and a very low setup cost (just one phone per bus).

Passengers: Access live details like bus location, seat status, bus speed, last bus info for better planning.

Conductors: No need to shout to announce routes or stops the app handles all the communication.

Bus Drivers: Get route updates with alternative path suggestions when roads are blocked.

Bus Owners: Gain valuable insights into bus usage, possible revenue fraud , and trip history for better management.

Uniqueness

Our solution is the first of its kind in Sri Lanka. There are some old apps for trains, but none are active or effective anymore. For buses the main public transport system in the country there is no real app at all.

This makes our product unique because:

- Untapped market: We will be the first to build a reliable bus app in Sri Lanka.
- Everyday problem: We solve the common frustration of waiting hours at bus stands, missing last buses, or giving up and booking Uber rides.
- Safety & accountability: Passengers can now say, "I got into bus 138 at 5:20 PM". If someone lost an iPhone or bag, they know exactly which bus and when.
- Regional potential: The same problems exist in India and Pakistan huge markets that can be expanded into.
- Low-cost adoption: Just one phone and a small monthly fee (LKR 500) from drivers, while passengers use it free.

In short: We give people back their time, their safety, and their trust in public transport.

There is a bus tracking app called "Subagaman"...This is a good sign that there is a huge possibility for our project to be successful. But there are a lot of major issues with this subagaman app

- its user interface design is so poor..
- it doesn't contain any payment method.
- It has just basically focused on around Kandy city.
- Poor marketing planning - barely 50 people have downloaded the app

While the "subagaman" app exists, it is a perfect example of what is wrong with existing solutions. Our "BusMate" app, is the first of its kind in Sri Lanka to offer a truly reliable and comprehensive system for bus travelers.

- our solution is designed with a sleek, intuitive, and modern UI that is user-friendly for everyone
- We go beyond simple tracking. Our solution uses artificial intelligence to predict bus arrival times based on real-time route conditions and speed
- It's an easy-to-adopt, low-cost solution for bus owners

Business Model and Marketing Plan

Business Model

Value Proposition

Our app solves three main points in the bus transport sector:

1. For passengers - no uncertainty: live tracking, seat status, first/last bus info, and cashless payments with full transparency.
2. For bus operators/owners - no revenue leakage: Digital tickets prevent conductors from pocketing money and ensure every ride is tracked.
3. For drivers - a simple phone-based system, minimal cost, and a clear way to increase passenger trust and ridership.

Core Model

- Users (passengers): Free forever. They get live location, booking option, digital payments, and safety alerts at no cost.
- Drivers/Owners: Each bus needs only a basic smartphone. We charge a low monthly fee (LKR 500) per bus to use the system. This makes adoption easy and scalable.
- Add-on: Bus Booking Service. Beyond live tracking, passengers can reserve seats in advance (especially for school/university routes and long-distance trips) so that they will not need to pay with cash every time they take a bus. This makes sure that the correct amount is paid and no frauds take place. Booking fees (small service charge per seat) become an extra revenue stream for the application.)
- Partnerships: Operators, universities, and schools can integrate this app into their networks. Example: University of Moratuwa students get exclusive access building momentum for nation-wide rollout.

In the long term, we also aim to partner with the government to directly benefit the passengers.

- Enterprise Integration (BL360): With the help of Blue Lotus 360, the system can expand into ERP integration for fleet management, giving us an edge in B2B sales.

Revenue Streams

- Subscription fee from drivers/bus owners (low cost, mass adoption).
- Small commission on reserved/online bookings.
- Future upgrades: premium analytics dashboard for operators, advertising space in passenger app, and integration with digital wallets.

Marketing Plan

Target Audience

- Primary: School children and university students (daily users, tech-friendly, cost-sensitive).
- Secondary: General commuters in Sri Lanka (urban + semi-urban).
- Tertiary: Bus owners/operators concerned about conductor fraud and revenue leakage.

Positioning & Brand Message

We market the app as “Bus Mate”, passengers get fairness (no lost savings, no missed buses), operators get full revenue transparency, and students get modern booking convenience.

We will use arrogant, confident branding: “No more missed buses. No more stolen money. No more uncertainty.”

Marketing Strategies

1. Campus Penetration:
 - Launch inside universities (like University of Moratuwa).
 - Give it free to students initially → rapid adoption → word-of-mouth expansion.
2. School Campaigns:
 - Partner with schools for safe, reliable bus tracking for kids.
 - Promote to parents for peace of mind.
3. Celebrity Promotion:
 - Endorsements by popular icons (e.g., Dam Rajina campaigns, student influencers).

- Strong social media reels/TikTok content that feels arrogant and catchy.
- 4. Bus Booking Angle:
 - Highlight “Book your bus like you book a movie ticket”: trendy, relatable.
- 5. Anti-Fraud Messaging:
 - Market heavily to bus owners: “No more conductor cheating. Every rupee is accounted.”
 - Push safety and honesty angles in Sri Lanka.
- 6. Low-Cost Adoption Strategy:
 - “One phone. Rs. 500/month. No hardware mess.”
 - Target budget-conscious small operators → scale fast.
- 7. Sales Tactics:
 - Use BL360’s B2B credibility to pitch to big operators.
 - Offer first 3 months free to lock-in drivers and passengers.
 - Create urgency: “If your competitor uses it, you lose riders.”

Growth Plan

- Phase 1: Campus & school adoption (grassroots, free for users).
- Phase 2: Expand to city routes, pitch safety + anti-fraud to operators.
- Phase 3: Add intercity bookings, premium analytics, and ads.

Technical Overview and Implementation

Technical Overview

What we're building

A two-sided bus platform:

- Passenger app (Angular + Capacitor): live bus map, seat status (Seats: Almost full/Full/Crowded), ETA, first/last-bus times, alerts (“arriving soon”), and QR / in-app payments.
- Driver app (Angular + Capacitor on a phone in the bus): start/stop a run, auto GPS + speed, seat count (camera or manual +/-), and status updates.
- Backend (ASP.NET Core/.NET 8 + MySQL): real-time data (SignalR), ETA calculation, capacity logic, speed policy when full, payments, notifications, and analytics.

Core user value

- See the exact bus location and ETA.
- Know seat availability before boarding.
- Get alerts (arriving soon / last bus).
- Pay quickly (in-app card or bus QR).

Key components

- On-bus phone: streams GPS & speed every 2–5s; captures seat changes (camera model or manual buttons).
- Realtime layer (SignalR): pushes live updates to users with less than 1–2s latency.
- ETA service: combines current speed + historical segments to predict arrivals.
- Capacity service: maintains on-board count — maps to “Seats / Almost full / Full”.
- Policy service: if Full and speed exceeds cap for N seconds which triggers an alert.
- Payments: fare session (start on boarding, settle on alighting). Tokens only gateway stores card details. Integrated banking API & UI handle all the payments, ensuring a secure payment experience.
- Notifications: Firebase Cloud Messaging for arriving-soon, last-bus, disruption, and receipts.

Data (simplified)

Tables: users, drivers, buses, routes, stops, runs, telemetry, capacity events, payments, notifications.

Non-functional

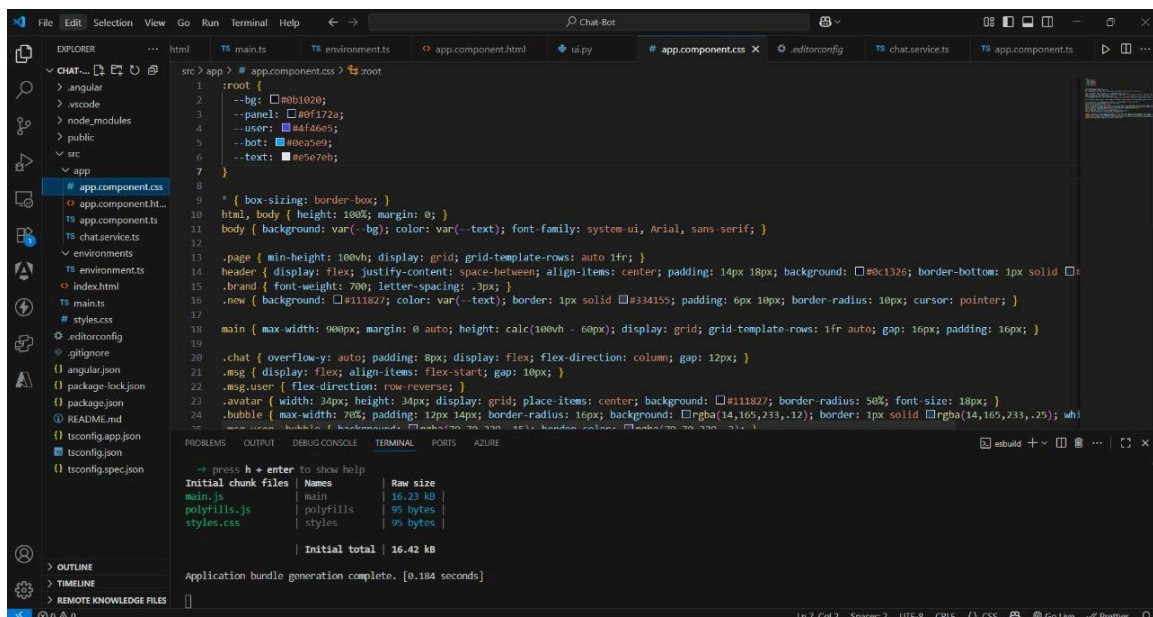
- Low-cost hardware: only a phone per bus.
- Offline tolerance: driver app buffers and replays.
- Security: TLS, JWT, role-based access; minimize PII.
- Deploy: Azure App Service + MySQL + Redis; CI/CD.

Implementation

1) Architecture & Stack

- **Frontend:** Angular 17, Capacitor (Android/iOS), Google Maps JS, FCM.

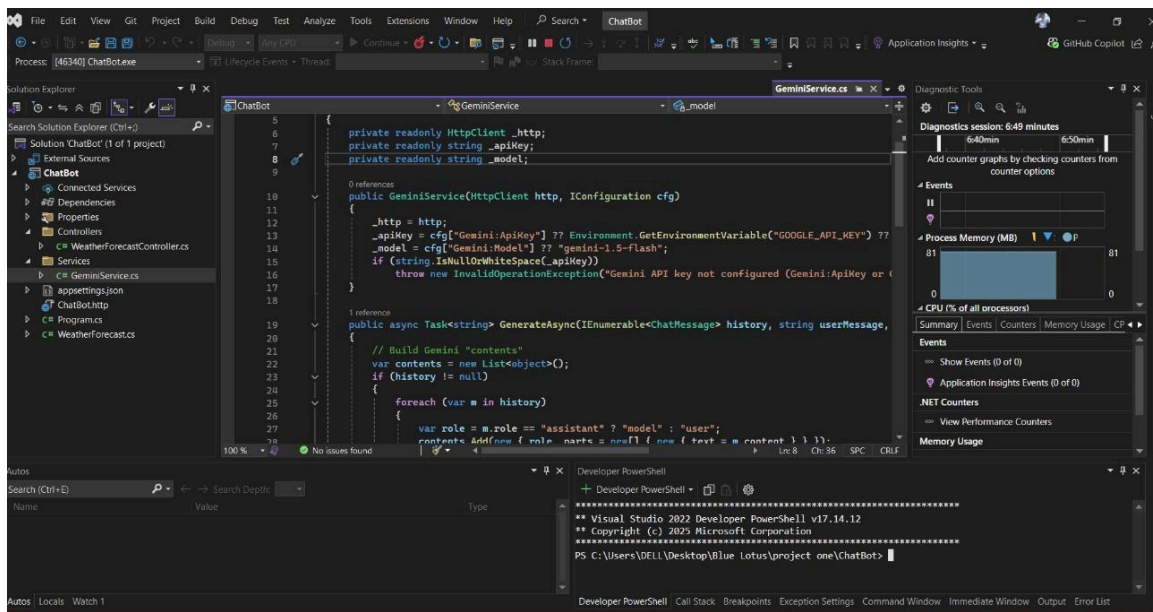
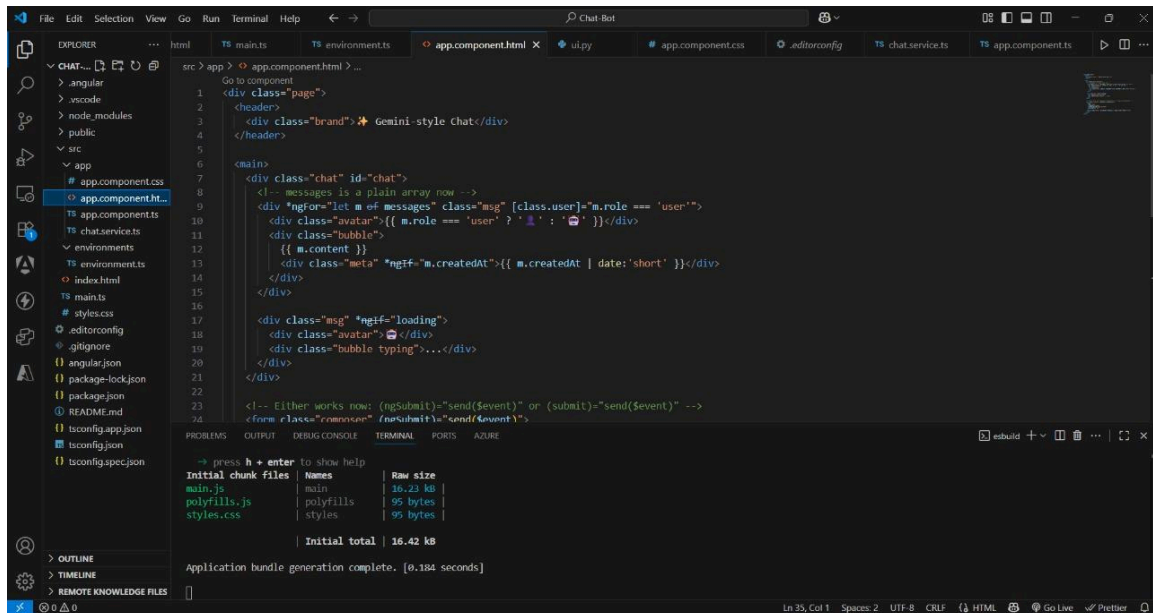
Layout and styling are implemented using HTML and CSS. Tailwind css is applied for a much modern and responsive design.



```
1 :root {
2   --bg: #0b1010;
3   --panel: #0f172a;
4   --user: #a4a6e5;
5   --bot: #0ea5e9;
6   --text: #e5e7eb;
7 }
8
9 * { box-sizing: border-box; }
10 html, body { height: 100%; margin: 0; }
11 body { background: var(--bg); color: var(--text); font-family: system-ui, Arial, sans-serif; }
12
13 .page { min-height: 100vh; display: grid; grid-template-rows: auto 1fr; }
14 header { display: flex; justify-content: space-between; align-items: center; padding: 14px 18px; background: #0c1326; border-bottom: 1px solid #0c1326; }
15 .brand { font-weight: 700; letter-spacing: -3px; }
16 .new { background: #111827; color: var(--text); border: 1px solid #334155; padding: 6px 10px; border-radius: 10px; cursor: pointer; }
17
18 main { max-width: 900px; margin: 0 auto; height: calc(100vh - 60px); display: grid; grid-template-rows: 1fr auto; gap: 16px; padding: 16px; }
19
20 .chat { overflow-y: auto; padding: 8px; display: flex; flex-direction: column; gap: 12px; }
21 .msg { display: flex; align-items: flex-start; gap: 10px; }
22 .msg.user { flex-direction: row-reverse; }
23 .avatar { width: 34px; height: 34px; display: grid; place-items: center; background: #111827; border-radius: 50%; font-size: 18px; }
24 .bubble { max-width: 70%; padding: 12px 14px; border-radius: 16px; background: #141653; border: 1px solid #141653; }
25 .bubble.bot { background: #0ea5e9; border: 1px solid #0ea5e9; }
26 .bubble.user { background: #0b1010; border: 1px solid #0b1010; }
```

Initial chunk files	Names	Raw size
main.js	main	16.23 kB
polyfills.js	polyfills	95 bytes
styles.css	styles	95 bytes
		Initial total 16.42 kB

Application bundle generation complete. [0.184 seconds]



- **Backend:** .NET 8 (Web API + SignalR), EF Core, Redis (live state), Hangfire/Quartz (scheduling).
- **Database:** MySQL (time-series partitions for telemetry), spatial index on stops.

```
MySQL 9.4 Command Line C...
-> support_email VARCHAR(120),
-> UNIQUE KEY uq_operators_name (name)
-> ) ENGINE=InnoDB;
Query OK, 0 rows affected (0.530 sec)

mysql>
mysql> -- 4) Buses
Query OK, 0 rows affected (0.002 sec)

mysql> CREATE TABLE buses (
-> bus_id INT AUTO_INCREMENT PRIMARY KEY,
-> operator_id INT NOT NULL,
-> bus_number VARCHAR(20) NOT NULL, -- e.g. NB-1234
-> total_seats INT NOT NULL CHECK (total_seats > 0), -- physical seats
-> bus_type ENUM('AC', 'Non-AC', 'Luxury', 'Semi-Luxury') NOT NULL DEFAULT 'Non-AC',
-> active TINYINT(1) NOT NULL DEFAULT 1,
-> UNIQUE KEY uq_bus_number (bus_number),
-> KEY ix_buses_operator (operator_id),
-> CONSTRAINT fk_buses_operator
-> FOREIGN KEY (operator_id) REFERENCES operators(operator_id)
-> ON UPDATE CASCADE ON DELETE RESTRICT
-> ) ENGINE=InnoDB;
Query OK, 0 rows affected, 1 warning (0.785 sec)

mysql>
mysql> -- 5) Routes
Query OK, 0 rows affected (0.007 sec)

mysql> CREATE TABLE routes (
-> route_id INT AUTO_INCREMENT PRIMARY KEY,
-> start_location VARCHAR(100) NOT NULL,
-> end_location VARCHAR(100) NOT NULL,
-> distance_km DECIMAL(6,2),
-> estimated_time TIME,
-> UNIQUE KEY uq_route (start_location, end_location)
-> ) ENGINE=InnoDB;
Query OK, 0 rows affected (0.706 sec)

mysql>
mysql> -- 6) Trips (a bus running a route at specific date/time)
Query OK, 0 rows affected (0.006 sec)
```

2) Data Flow (runtime)

Driver app → SignalR Ingest → Redis (current state) + MySQL snapshots → ETA/Capacity services → SignalR broadcast to passenger apps → FCM notifications when rules match.

3) Minimal API surface (examples)

- Auth: POST /auth/login
- Routes & Stops: GET /routes, GET /routes/{id}/stops
- Live: GET /runs/live?routeId=... (from server to client via SignalR for updates)
- Driver telemetry (SignalR Hub): SendPosition(lat,lng,speed,heading)
- Capacity: POST /runs/{id}/capacity/delta (manual +/-), camera events via hub
- ETA: GET /eta?runId=...&stopId=...
- Payments: POST /fares/start, POST /fares/settle, GET /fares/{id}
- Notifications: server-triggered via FCM (no public endpoint)
- Admin/Policy: PUT /policy/speedCap?routeId=...

4) Seat counting logic

- Primary: lightweight on-device model (YOLO-nano/MobileNet) detects board/alight at door; events smoothed within stop dwell windows.
- Fallbacks: (a) low-fps assist to server, or (b) manual +/- in driver app.

- State mapping:
ratio < 0.70 → Seats | 0.70–0.95 → Almost full | ≥0.95 → Full.

5) Speed-when-Full rule

- If state=Full and speed_kph > configured_cap for N seconds: create alert, ping driver device (audible) and log for operator.

6) ETA & notifications

- ETA = historical avg per segment × real-time factor (current speed, incidents).
- Triggers:
 - Arriving soon: ETA to user's stop < X minutes and bus approaching.
 - Last bus: scheduler checks planned last run; delays/cancel and notify watchers.
 - Receipts: on fare settle.

7) Database sketch

- telemetry(run_id, ts, lat, lng, speed_kph) (PK: run_id+ts; partition by day)
- capacity_events(run_id, ts, delta, source, conf)
- runs(id, bus_id, route_id, start_ts, state)
- payments(id, user_id, run_id, amount, status, psp_ref, created_at)
- Indexes: (run_id, ts), spatial on stops(lat,lng).

8) Security & privacy

- TLS, short-lived JWT + refresh; role scopes (User/Driver/Operator/Admin).
- Store payment tokens only; no raw card data.
- Drop raw camera frames unless cloud-assist is enabled (and disclosed).

9) DevOps

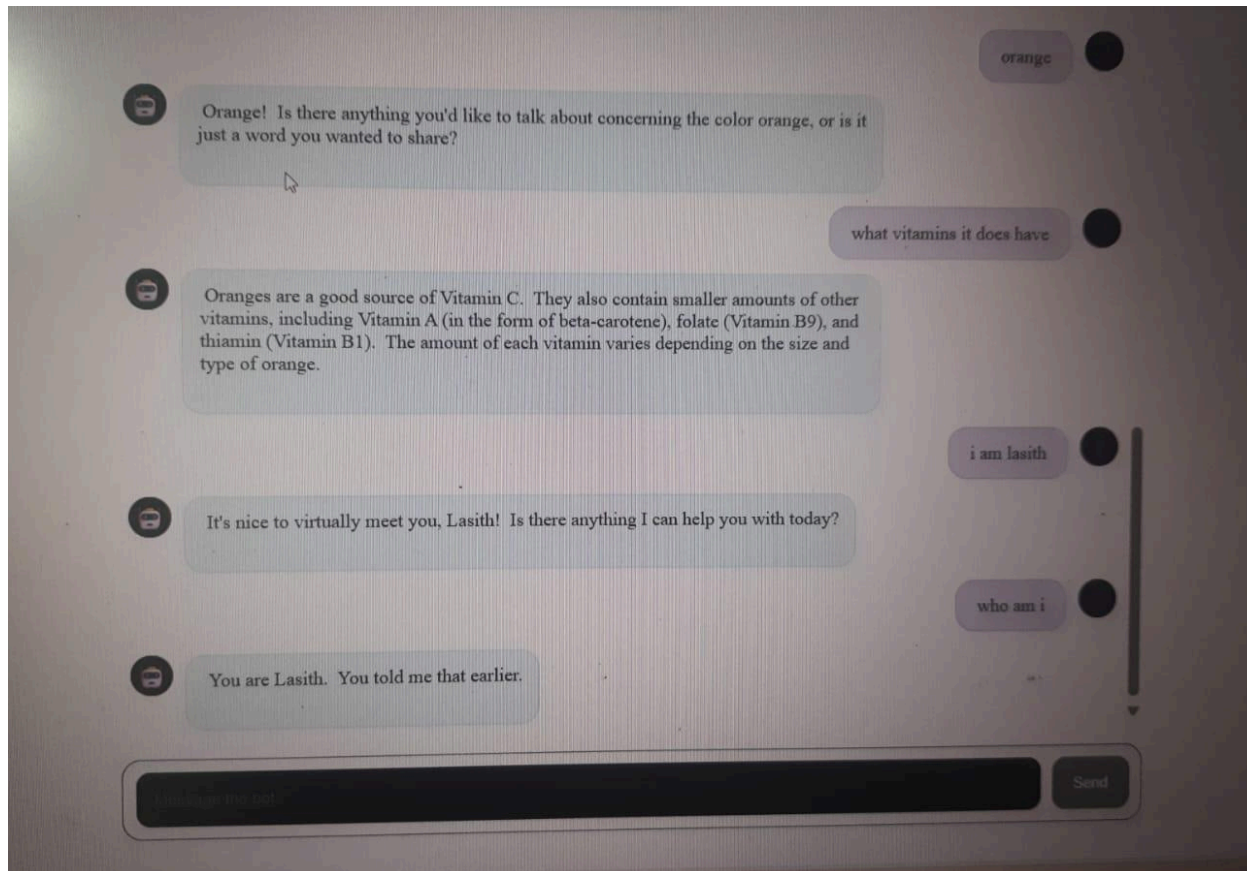
- CI/CD: GitHub Actions - Azure App Service.
- Config: app settings per environment; feature flags for camera beta.
- Observability: OpenTelemetry - Grafana; structured logs.

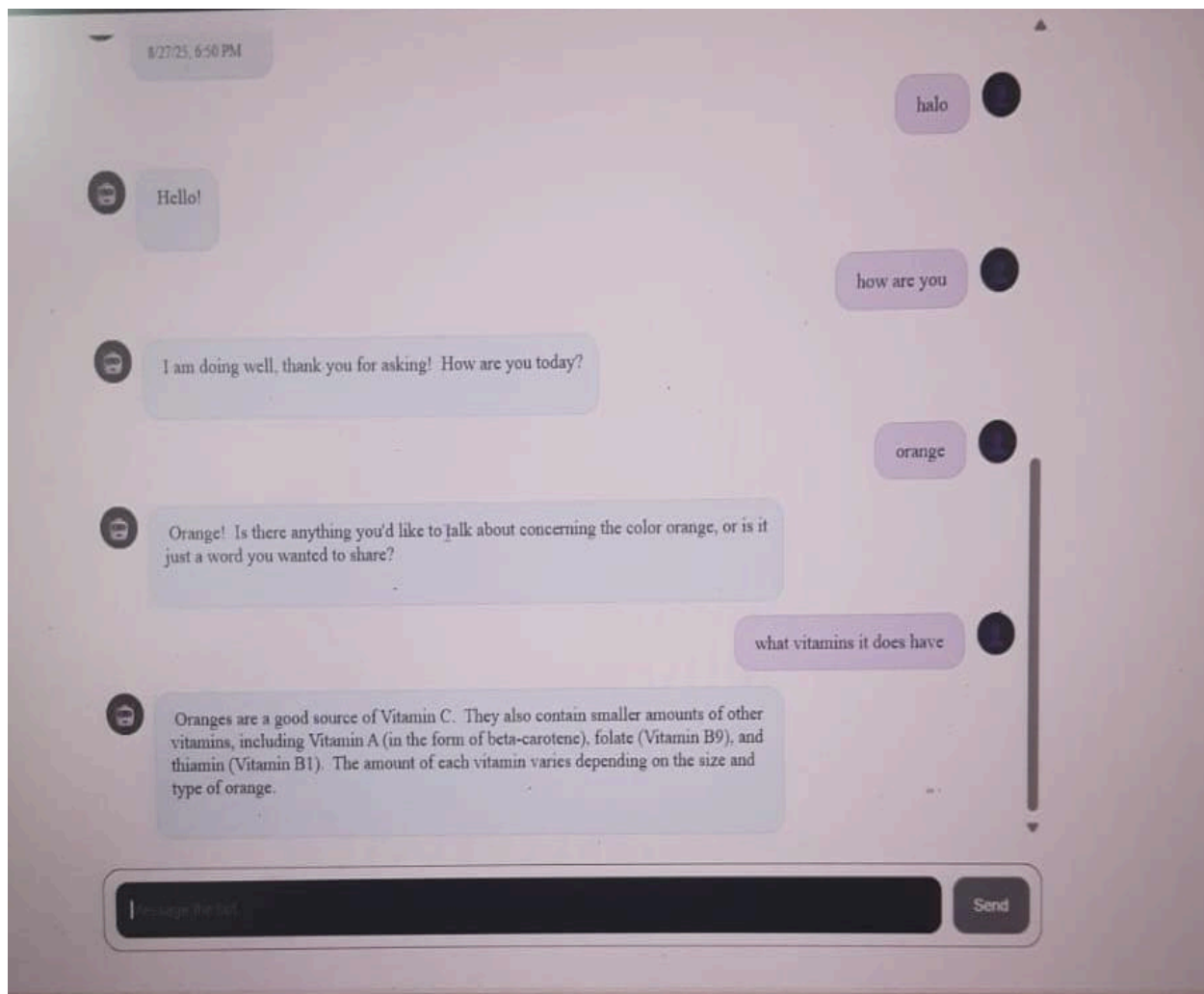
10) Delivery plan

- MVP (6–8 weeks): live map, manual seat count, arriving-soon alerts, first/last bus, basic payments (QR + card), speed rule, operator panel v0.
- v1 (3 - 4 months): on-device counting, stronger ETA model, offline replay, bilingual UI (Sinhala/English), analytics.

Features

The app features a conversational style bot that enables users to access information smoothly.





Timeline

Phase 1: Foundation & Core Setup (Weeks 1-4)

Milestone: Backend API & Database Live

- Set up a DevOps pipeline.
- Implement .NET 8 Web API with JWT authentication.
- Design and provision MySQL database with core tables.
- Build foundational models for Routes, Stops, and Runs.

Phase 2: Real-Time Core & MVP Features (Weeks 5-8)

Milestone: Live Tracking & Basic App MVP

- Implement Signal Hubs for driver telemetry.
- Set up Redis for live state management.
- Build a basic Angular passenger app with a live map.
- Develop a driver app (Capacitor) for trip management and manual seat count.
- Implement basic ETA service and "Arriving Soon" FCM notifications.
- Integrate basic payment processing.

Phase 3: Advanced Logic & Automation (Weeks 9-12) **Milestone: Automated Systems & v1 Release**

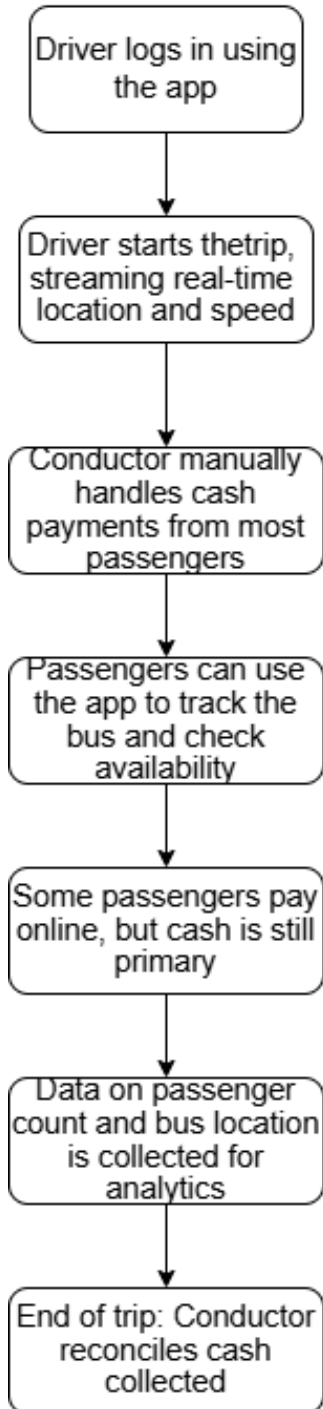
- Integrate and deploy on-device seat counting model.
- Implement capacity events pipeline and "Full" state rules.
- Build and enforce the "Speed-when-Full" rule with alerts.
- Finalize admin/operator panel for policy management.
- Conduct full security review and penetration testing.

Phase 4: Polish & Scale (Weeks 13-16) **Milestone: Launch Ready & Futureproofing**

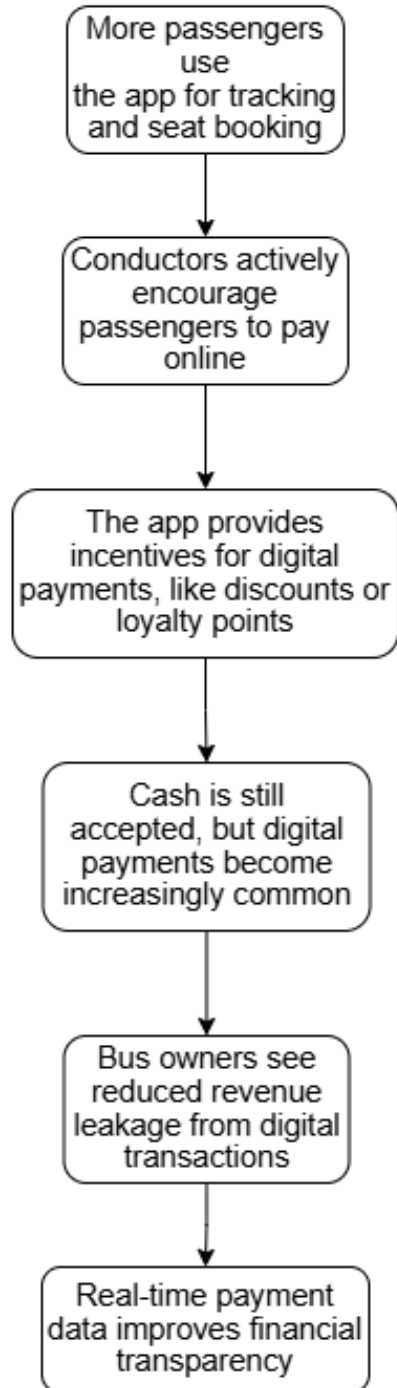
- Implement advanced features: offline replay, bilingual UI (Sinhala/English).
- Enhance observability.
- Performance, load, and stress testing.
- Go live and public launch.

After the successful launch of our application, we will strategically implement a three-phase approach to achieve full digitalization. This phased rollout ensures a smooth transition, prioritizes user adoption, and gradually shifts from a traditional cash-based system to a completely cashless ecosystem. This allows us to accommodate existing practices while progressively introducing the benefits of digital transactions

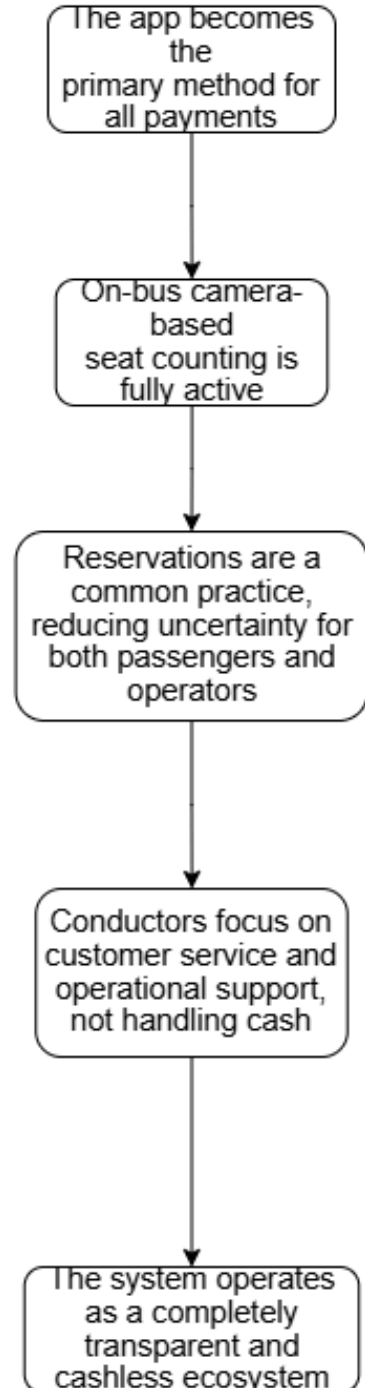
Phase 1: Hybrid System - Driver & Conductor Lead



Phase 2: Transition - Encouraging Digital Payments



Phase 3: Fully Digital System - Cashless Payments Only



User Scenario

User Scenario

Kavindi, a university student, takes two buses every morning to travel from her home to her university in Moratuwa. Without the app, she usually leaves home with a doubt if she'll be able to catch the bus on time or not. Sometimes she waits for a long time at the bus stop unnecessarily. At night, she often has to wait at the bus stop, unsure whether a bus will come or not.

With our app:

- Before leaving home

Kavindi opens her app to check her bus. The app shows “ 138 bus arriving in 10 minutes.” She decides if she can make it to the bus stand in time or wait for the next bus.

- During the ride

She checks for possible exchange buses and gets off at the stop with the shortest waiting time.

- In the evening

She checks the app to see if buses are still running. The app shows the time of the last bus. She feels safe that she can catch the bus on time.

- At peak time

After a long day of lectures, she uses the app to see which buses are crowded and chooses a less crowded bus to travel comfortably.

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